Q # 01 Apply nth-term test

i) $\sum_{n=1}^{\infty} n \sin\frac{1}{n}$

Let $a_n = n \sin\frac{1}{n}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} n.\sin\frac{1}{n}$ $(\infty \times 0)$

$= \lim_{n\to\infty} \frac{\sin\frac{1}{n}}{1/n}$ $(0/0)$

using L'Hopital's Rule

$= \lim_{n\to\infty} \frac{\cos 1/n \cdot -1/n^2}{\cancel{-1/n^2}}$

$= \lim_{n\to\infty} \cos 1/n = \cos 0 = 1$

$\Rightarrow \lim_{n\to\infty} a_n = 1 \neq 0$

$\Rightarrow$ Given Series diverges by nth-term test.

ii) $\sum_{n=2}^{\infty} \frac{\ln(n^2)}{n}$

Let $a_n = \frac{\ln(n^2)}{n}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{\ln(n^2)}{n}$

$= \lim_{n\to\infty} \frac{2\ln(n)}{n}$

$= 2 \lim_{n\to\infty} \frac{\ln(n)}{n}$ (∞/∞)

$$\lim_{n\to\infty} a_n = 2 \lim_{n\to\infty} \frac{1/n}{1}$$

$$= 2 \cdot 0$$

$$\lim_{n\to\infty} a_n = 0$$

$\Rightarrow$ nth-term test fail.

iii) $$\sum_{n=1}^{\infty} \frac{5^n + (-1)^n}{5^{n+1} + (-1)^{n+1}}$$

Let $$a_n = \frac{5^n + (-1)^n}{5^{n+1} + (-1)^{n+1}} = \frac{\cancel{5^n}\left[1 + (-1/5)^n\right]}{\cancel{5^n}\left[5^1 + (-1/5)^n \cdot (-1)^1\right]}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1 + (-1/5)^n}{5 + (-1/5)^n (-1)}$$

$\because \lim_{n\to\infty} x^n = 0$

$; |x| < 1$

$\Rightarrow \lim_{n\to\infty} (-1/5)^n = 0$

$$= \lim_{n\to\infty} \frac{1 + 0}{5 + 0 \cdot (-1)}$$

$$\lim_{n\to\infty} a_n = \frac{1}{5} \neq 0$$

$\Rightarrow$ Given series diverges by nth-term test.

iv) $$\sum_{n=2}^{\infty} \frac{(n-4)!}{(n-1)!}$$

Let $$a_n = \frac{(n-4)!}{(n-1)!} = \frac{\cancel{(n-4)!}\cancel{(n-3)(n-2)(n-1)}!}{(n-1)(n-2)\cdot(n-3)\cdot\cancel{(n-4)!}}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1}{(n-1)(n-2)(n-3)}$$ $\therefore \left(\frac{1}{\infty}=0\right)$

$$= 0$$

~~Given Series diverges by~~ nth-term Test Fail.

v) $$\sum_{n=1}^{\infty} \frac{2^n}{(2n)!}$$

Let $$a_n = \frac{2^n}{(2n)!}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{2^n}{(2n)!}$$

$$= \lim_{n\to\infty} \left[\frac{1}{(2n)(2n-1)\cdots(2n-(n+1))} \cdot \frac{2^n}{n!}\right]$$

$$= \lim_{n\to\infty} \frac{1}{(2n)(2n-1)\cdots(2n-(n+1))} \cdot \lim_{n\to\infty} \frac{2^n}{n!}$$

$$= 0 \cdot 0$$ $\therefore \lim_{n\to\infty} \frac{x^n}{n!} = 0$

$$= 0$$ $\Rightarrow$ nth-Term Test Fail.

$$(2n)! = (2n)\cdot(2n-1)\cdot(2n-2)\cdots(2n-(n+1))\cdot(2n-n)$$

Q#02 Find the Sum of given Series, if possible.

2) $$\sum_{n=1}^{\infty} \frac{1}{9n^2+3n-2}$$

$9n^2+3n-2$
$= 9n^2-3n+6n-2$
$= (3n-1)(3n+2)$

$$= \sum_{n=1}^{\infty} \frac{1}{(3n-1)(3n+2)}$$

$$\frac{1}{(3n-1)(3n+2)} = \frac{A}{3n-1} + \frac{B}{3n+2}$$

$\Rightarrow A = 1/3$
$B = -1/3$

$$= \sum_{n=1}^{\infty} \left[\frac{A}{(3n-1)} + \frac{B}{(3n+2)} \right]$$

$$= \sum_{n=1}^{\infty} \frac{1/3}{(3n-1)} - \frac{1/3}{(3n+2)}$$

$$= \frac{1}{3} \sum_{n=1}^{\infty} \left[\frac{1}{(3n-1)} - \frac{1}{(3n+2)} \right]$$

$$= \frac{1}{3} \left[\left(\frac{1}{2} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{8}\right) + \left(\frac{1}{8} - \frac{1}{11}\right) + \cdots + \left(\frac{1}{(3n-1)} - \frac{1}{(3n+2)}\right) + \cdots \right.$$

Let

$$S_n = \frac{1}{3} \left[\left(\frac{1}{2} - \cancel{\frac{1}{5}}\right) + \left(\cancel{\frac{1}{5}} - \cancel{\frac{1}{8}}\right) + \left(\cancel{\frac{1}{8}} - \cancel{\frac{1}{11}}\right) + \cdots \left(\cancel{\frac{1}{3n-1}} - \frac{1}{3n+2}\right) \right.$$

$$S_n = \frac{1}{3} \left[\frac{1}{2} - \frac{1}{3n+2} \right]$$

$$\lim_{n \to \infty} S_n = \frac{1}{3} \left[\frac{1}{2} - \lim_{n \to \infty} \frac{1}{3n+2} \right]$$

$$\lim_{n\to\infty} S_n = \frac{1}{3}\left[\frac{1}{2} - 0\right] = \frac{1}{6}$$

$\Rightarrow$ The Sequence of Partial Sums $\{S_n\}$ Converges hence the given Series also Converges and

$$\sum_{n=1}^{\infty} \frac{1}{9n^2+3n-2} = \frac{1}{6}.$$

ii) $\displaystyle\sum_{n=1}^{\infty} \ln\left(\frac{n}{n+1}\right)$

$$= \sum_{n=1}^{\infty} \left[\ln(n) - \ln(n+1)\right]$$

Let

$$S_n = [\ln(1) - \cancel{\ln(2)}] + [\cancel{\ln(2)} - \cancel{\ln(3)}] + [\cancel{\ln(3)} - \ln(4)] + \cdots + [\cancel{\ln(n)} - \ln(n+1)].$$

$$S_n = \ln(1) - \ln(n+1)$$

$\because \ln(1) = 0$

$$\lim_{n\to\infty} S_n = -\lim_{n\to\infty} \ln(n+1)$$

$$= -\infty.$$

$\Rightarrow$ The Sequence of Partial Sums $\{S_n\}$ diverges. Hence given series also diverges and its Sum is not Possible.

iii) $\sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n^2+n}}$

$$= \sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n(n+1)}}$$

$$= \sum_{n=1}^{\infty} \frac{\cancel{\sqrt{n+1}}}{\sqrt{n}\,\cancel{\sqrt{n+1}}} - \frac{\cancel{\sqrt{n}}}{\cancel{\sqrt{n}}\cdot\sqrt{n+1}}$$

$$= \sum_{n=1}^{\infty} \left[\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right]$$

Let $S_n = \left[\frac{1}{\sqrt{1}} - \cancel{\frac{1}{\sqrt{2}}}\right] + \left[\cancel{\frac{1}{\sqrt{2}}} - \cancel{\frac{1}{\sqrt{3}}}\right] + \left[\cancel{\frac{1}{\sqrt{3}}} - \cancel{\frac{1}{\sqrt{4}}}\right] + \cdots + \left[\cancel{\frac{1}{\sqrt{n}}} - \frac{1}{\sqrt{n+1}}\right]$

$$S_n = 1 - \frac{1}{\sqrt{n+1}}$$

$$\lim_{n\to\infty} S_n = 1 - \lim_{n\to\infty} \frac{1}{\sqrt{n+1}}$$

$$= 1$$

$\Rightarrow$ The Sequence of Partial Sums $\{S_n\}$ Converges.
Hence the given series also Converges and its Sum is 1. i.e

$$\sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n^2+n}} = 1$$

iv) $$\sum_{n=1}^{\infty} \frac{40n}{(2n-1)^2(2n+1)^2} \quad \text{— (A)}$$

$$\frac{40n}{(2n-1)^2(2n+1)^2} = \frac{A}{(2n-1)} + \frac{B}{(2n-1)^2} + \frac{C}{(2n+1)} + \frac{D}{(2n+1)^2}$$

$$40n = A(2n-1)(2n+1)^2 + B(2n+1)^2 + C(2n-1)^2(2n+1) + D(2n-1)^2$$

$\Rightarrow$ $A = 0$, $C = 0$

$B = 5$, $D = -5$

So, we can write the given series as

$$= \sum_{n=1}^{\infty} \left[\frac{5}{(2n-1)^2} - \frac{5}{(2n+1)^2} \right]$$

$$= 5 \sum_{n=1}^{\infty} \left[\frac{1}{(2n-1)^2} - \frac{1}{(2n+1)^2} \right]$$

Let $S_n = 5\left[\left(1 - \cancel{\frac{1}{9}}\right) + \left(\cancel{\frac{1}{9}} - \cancel{\frac{1}{25}}\right) + \left(\cancel{\frac{1}{25}} - \cancel{\frac{1}{49}}\right) + \cdots + \left(\cancel{\frac{1}{(2n-1)^2}} - \frac{1}{(2n+1)^2}\right)\right.$

$$S_n = 5\left[1 - \frac{1}{(2n+1)^2}\right]$$

$$\lim_{n\to\infty} S_n = 5\left[1 - \lim_{n\to\infty} \frac{1}{(2n+1)^2}\right]$$

$$\lim_{n\to\infty} S_n = 5[1-0]$$

$$= 5$$

$\Rightarrow$ The sequence of partial sums $\{S_n\}$ converges. Hence the given series also converges and its sum is 5. i.e

$$\sum_{n=1}^{\infty} \frac{40n}{(2n-1)^2(2n+1)^2} = 5.$$

v) $$(1+1)+\left(\frac{1}{2}+\frac{1}{2^{2/3}}\right)+\left(\frac{1}{4}+\frac{1}{3^{2/3}}\right)+\left(\frac{1}{8}+\frac{1}{4^{2/3}}\right)+\cdots$$

OR

$$\left(1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots\right)+\left(\frac{1}{1^{2/3}}+\frac{1}{2^{2/3}}+\frac{1}{3^{2/3}}+\frac{1}{4^{2/3}}+\cdots\right)$$

Convergent Geometric Series with $r = 1/2$

It is divergent p-series with $p = 2/3 < 1$.

OR

$$\sum_{n=0}^{\infty} \frac{1}{2^n} + \sum_{n=1}^{\infty} \frac{1}{n^{2/3}}$$

Now the sum of given series is not possible because $\sum_{n=1}^{\infty} \frac{1}{n^{2/3}}$ diverges.

vi) $\sum_{n=1}^{\infty} \frac{e^{2n} + e^{-2n}}{e^{n} + e^{-n}}$

Let $a_n = \frac{e^{2n} + e^{-2n}}{e^{n} + e^{-n}}$

$$= \frac{e^{2n}(1 + e^{-4n})}{e^{n}(1 + e^{-2n})}$$

$$= e^{n}\left(\frac{1 + e^{-4n}}{1 + e^{-2n}}\right)$$

$$\lim_{n \to \infty} a_n = \lim_{n \to \infty} e^{n} \cdot \lim_{n \to \infty} \frac{\left(1 + \frac{1}{e^{4n}}\right)}{\left(1 + \frac{1}{e^{2n}}\right)}$$

$\because \lim x^n = \infty$; $x > 1$

$\Rightarrow \lim_{n \to \infty} e^n = \infty$ and $\lim_{n \to \infty} \frac{1}{e^{4n}} = 0$

$\lim_{n \to \infty} \frac{1}{e^{2n}} = 0$

$\Rightarrow$

$$\lim_{n \to \infty} a_n = \infty \cdot \frac{(1+0)}{(1+0)}$$

$$= \infty \cdot 1$$

$$= \infty$$

Its Sum is not possible.

$\Rightarrow$ Series diverges by nth-term Test.

Q#3 Apply Integral Test

i) $\sum_{n=1}^{\infty} n e^{-n^2}$

let $a_n = n e^{-n^2}$

i) a_n is +ve $\forall\ n \geq 1$

ii) a_n is Continuous $\forall\ n \geq 1$

iii) $a_n = n e^{-n^2}$

$a_n' = n(-2n e^{-n^2}) + e^{-n^2}$

$= e^{-n^2}(1-2n^2) < 0 \quad \forall\ n \geq 1$

$\Rightarrow$ a_n is also decreasing function.

Now apply integral Test

$$\int_1^{\infty} f(n)\,dn = \lim_{b\to\infty} \int_1^{b} n e^{-n^2}.dn$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$= -\frac{1}{2} \lim_{b\to\infty} \int_1^{b} -2n e^{-n^2} dn$$

$$\cancel{\because \int f'(x)[f(x)]^n = \frac{[f(x)]^{n+1}}{n+1} + C}$$

$$= -\frac{1}{2} \lim_{b\to\infty} \frac{(e^{-n^2})^{\cancel{2}}}{\cancel{2}} \Bigg|_1^b$$

$$= -\frac{1}{2} \lim_{b\to\infty} e^{-n^2} \Big|_1^b$$

$$= -\frac{1}{2} \lim_{b\to\infty} \left[\frac{1}{e^{n^2}} \Big|_1^b \right]$$

$$= -\frac{1}{9} \lim_{b \to \infty} \left[\frac{1}{e^{b^2}} - \frac{1}{e^{1^2}} \right]$$

$$= -\frac{1}{9} \left[\lim_{b \to \infty} \frac{1}{e^{b^2}} - \frac{1}{e^{1}} \right] \qquad \because \lim_{b \to \infty} \frac{1}{e^{b^2}} = 0$$

$$= -\frac{1}{9} \left[0 - \frac{1}{e^{1}} \right]$$

$$= \frac{1}{9e}$$

$\Rightarrow$ The integral $\int_1^\infty n e^{-n^2} dn$ Converges,

hence given series also Converges.

ii) $\displaystyle\sum_{n=1}^{\infty} \frac{e^n}{1+e^{2n}}$

Let $a_n = \dfrac{e^n}{1+e^{2n}}$

1) a_n is +ve $\forall\ n \geq 1$.

2) a_n is Continuous $\forall\ n \geq 1$.

3) $a_n = \dfrac{e^n}{1+e^{2n}}$

$$a_n' = \frac{(1+e^{2n}) \cdot e^n - e^n \cdot 2e^{2n}}{(1+e^{2n})^2} = \frac{e^n[1+e^{2n} - 2e^{2n}]}{(1+e^{2n})^2}$$

$$a_n' = \frac{e^n[1-e^{2n}]}{(1+e^{2n})^2} < 0 \quad \forall\ n \geq 1$$

$\Rightarrow$ a_n is a decreasing function.

$$\int_N^{\infty} \frac{e^n}{1+e^{2n}} dn = \int_1^{\infty} \frac{e^n}{1+e^{2n}} \cdot dn$$

Let $u = e^n$
$du = e^n dn$

$$= \int_e^{\infty} \frac{1}{1+u^2} \cdot du$$

$$= \lim_{b\to\infty} \int_e^b \frac{1}{1+u^2} \cdot du$$

$$= \lim_{b\to\infty} \operatorname{Tan}^{-1} u \Big|_e^b$$

$$= \operatorname{Tan}^{-1}(\infty) - \operatorname{Tan}^{-1}(e) = \pi/2 - \operatorname{Tan}^{-1}(e)$$
$$= 0.35$$

So the Series Converges by Integral Test.

iii) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}$

Let $a_n = \frac{1}{\sqrt{n}(\sqrt{n}+1)}$

1) a_n is +ve $\forall$ $n \geq 1$.

2) a_n is Continuous $\forall$ $n \geq 1$

3) $$a_n' = \frac{\sqrt{n}(\sqrt{n}+1).0 - 1\frac{d}{dn}\left(\sqrt{n}(\sqrt{n}+1)\right)}{\left[\sqrt{n}(\sqrt{n}+1)\right]^2}$$

$$= \frac{-\frac{1}{2\sqrt{n}}(2\sqrt{n}+1)}{\left[\sqrt{n}(\sqrt{n}+1)\right]^2} < 0 \quad \forall\ n \geq 1$$

$\Rightarrow$ a_n is a decreasing function.

$$\int_N^{\infty} f(n)\,dn = \int_1^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}\,dn$$

$$= 2\int_2^{\infty} \frac{1}{U}\,du$$

Let $U = \sqrt{n}+1$

$du = \frac{1}{2\sqrt{n}}\,dn$

$2du = \frac{1}{\sqrt{n}}\,dn$

when $n=1$

$\Rightarrow U = 2$

when $n \to \infty$

$U \to \infty$

$$= 2 \lim_{b \to \infty} \int_2^b \frac{du}{u}$$

$$= 2 \lim_{b \to \infty} \ln(u) \Big|_2^b$$

$$= 2 \lim_{b \to \infty} [\ln(b) - \ln(2)]$$

$$= 2 [\ln(\infty) - \ln(2)]$$

$$= \infty$$

$\Rightarrow$ Series diverges by Integral Test.

iv) $\displaystyle\sum_{n=1}^{\infty} \frac{\ln(n)}{\sqrt{n}}$

let $a_n = \dfrac{\ln(n)}{\sqrt{n}}$

1) a_n is +ve $\forall\ n \geq 1$.

2) a_n is Continuous $\forall\ n \geq 1$.

3) a_n is decreasing $\forall\ n \geq 2$.

$$\int_{N}^{\infty} f(n)\,dn = \int_{2}^{\infty} \frac{\ln n}{\sqrt{n}}\,dn$$

Let $U = \ln n$

$du = \frac{1}{n}\,dn$

$= \frac{1}{\sqrt{n}.\sqrt{n}}\,dn$

$\sqrt{n}\,du = \frac{1}{\sqrt{n}}\,dn$

$\because U = \ln n$

$e^{U} = n$

$e^{U/2} = \sqrt{n}$

$e^{U/2}\,du = \frac{1}{\sqrt{n}}\,dn$

$U = \ln n$

when $n = 2$

$\Rightarrow U = \ln(2)$

when $U \to \infty$

$U \to \infty$

$$\int_{2}^{\infty} \frac{\ln n}{\sqrt{n}}\,dn = \int_{\ln(2)}^{\infty} e^{U/2}.U\,du$$

$$= \lim_{b\to\infty} \int_{\ln(2)}^{b} U.e^{U/2}\,du.$$

(integration by parts)

$$= U\frac{e^{U/2}}{1/2} - \int \frac{e^{U/2}}{1/2}.1\,du$$

$$= 2\left[U e^{U/2} - \int e^{U/2}\,du\right]$$

$$= \lim_{b\to\infty} 2\left[U e^{U/2} - 2e^{U/2}\right]\Big|_{\ln(2)}^{b}$$

$$= 2 \lim_{b\to\infty} e^{U/2}(U-2)\Big|_{\ln(2)}^{b}$$

$$= 2 \lim_{b\to\infty} \left[e^{b/2}(b-2) - 2e^{\frac{\ln(2)}{2}}(\ln 2 - 2) \right]$$

$$= \infty$$

So, the given series diverges.